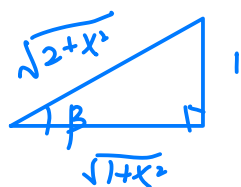
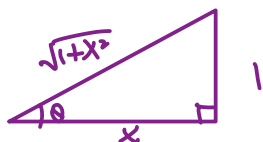


1. (Stewart, Chapter 5 Review, #39 and #92)

(a) Show that



$$\cos(\arctan(\sin(\operatorname{arccot} x))) = \sqrt{\frac{x^2+1}{x^2+2}}$$

$$\theta = \operatorname{arccot} x \Rightarrow \cot \theta = x$$

$$= \cot^{-1}(x)$$

$$\sin \theta = \frac{1}{\sqrt{1+x^2}}$$

$$\arctan\left(\frac{1}{\sqrt{1+x^2}}\right)$$

$$\beta$$

$$\tan \beta = \frac{1}{\sqrt{1+x^2}}$$

$$\cos \beta = \frac{\sqrt{1+x^2}}{\sqrt{2+x^2}}$$

(b) Differentiate $y = \ln(\cosh 3x)$.

$$\frac{dy}{dx} = (\ln |\cosh 3x|)'$$

$$\stackrel{\text{chain rule}}{=} \frac{1}{\cosh 3x} \cdot (\cosh 3x)'$$

$$\stackrel{\text{chain rule}}{=} \frac{1}{\cosh 3x} \cdot \sinh 3x \cdot 3$$

Recall:

$$\cosh(x) = \frac{e^x + e^{-x}}{2}$$

$$\sinh(x) = \frac{e^x - e^{-x}}{2}$$

$$= 3 \tanh 3x$$

$$\Rightarrow (\cosh(x))' = \sinh(x)$$

$$(\sinh(x))' = \cosh(x)$$

2. (Stewart, Chapter 5 Review, #27 and #88)

(a) Evaluate the integral $\int \frac{x}{\sqrt{1-x^4}} dx$.

Sol: Let $u = x^2 \Rightarrow du = 2x dx$

$$\int \frac{x}{\sqrt{1-x^4}} dx = \frac{1}{2} \int \frac{du}{\sqrt{1-u^2}}$$

$$= \frac{1}{2} \arcsin u + C$$

$$= \frac{1}{2} \arcsin(x^2) + C$$

(b) Differentiate $y = x \sinh(x^2)$

$$y' = (x \sinh(x^2))'$$

$$\stackrel{\text{chain}}{=} (x)' \sinh(x^2) + x (\sinh(x^2))'$$

$$= \sinh(x^2) + x \cosh(x^2) \cdot \underline{2x}$$

$$= \sinh(x^2) + 2x^2 \cosh(x^2)$$

3. (Stewart, Chapter 5 Review, #41 and #82)

(a) Evaluate the integral $\int_0^{\pi/2} \frac{\cos x}{1 + \sin^2 x} dx$.

$$\begin{aligned} \text{Let } u &= \sin x & \int_0^1 \frac{du}{1+u^2} \\ du &= \cos x dx & = \arctan u \Big|_0^1 \\ & & = \arctan 1 - \arctan 0 \\ & & = \frac{\pi}{4} - 0 = \frac{\pi}{4}. \end{aligned}$$

(b) Differentiate $y = \cosh^{-1}(\sinh x)$

$$\begin{aligned} y' &= (\cosh^{-1}(\sinh(x)))' \\ \text{chain} & & \text{Recall } (\cosh^{-1}(x))' = \frac{1}{\sqrt{x^2-1}} \\ &= \frac{1}{\sqrt{(\sinh(x))^2-1}} \cdot (\sinh(x))' \\ &= \frac{1}{\sqrt{(\sinh(x))^2-1}} \cdot \cosh(x). \end{aligned}$$

4. (Stewart, Chapter 5 Review, #64, and §5.7, #61)

(a) Evaluate the limit $\lim_{x \rightarrow \infty} \arctan(x^3 - x)$.

$$= \arctan(+\infty)$$

$$= \frac{\pi}{2}.$$

(b) Evaluate the integral $\int \frac{e^x}{1 - e^{2x}} dx$.

$$\text{Let } u = e^x \\ du = e^x dx$$

$$= \int \frac{du}{1 - u^2}$$

$$= \tanh^{-1}(u) + C$$

$$= \tanh^{-1}(e^x) + C.$$

$$\tanh^{-1}(x) = \frac{1}{2} \ln \left(\frac{1+x}{1-x} \right)$$

$$= \frac{1}{2} \ln \left(\frac{1+e^x}{1-e^{-x}} \right) + C.$$

Recall:

$$(\tanh^{-1}(x))' = \frac{1}{1-x^2}$$

5. (Stewart, Chapter 5 Review, #81, and §5.7, #43)

(a) Differentiate $y = \arctan(\arcsin \sqrt{x})$.

$$\begin{aligned}
 y' &= (\arctan(\arcsin \sqrt{x}))' \\
 &\stackrel{\text{chain}}{=} \frac{1}{1+(\arcsin \sqrt{x})^2} \cdot (\arcsin \sqrt{x})' \\
 &\stackrel{\text{Recall:}}{=} \frac{1}{1+(\arcsin \sqrt{x})^2} \cdot \frac{1}{\sqrt{1-(\sqrt{x})^2}} \cdot (\sqrt{x})' \\
 &= \frac{1}{1+(\arcsin \sqrt{x})^2} \cdot \frac{1}{\sqrt{1-x}} \cdot \frac{1}{2\sqrt{x}}
 \end{aligned}$$

(b) Show that $\frac{d}{dx} \arctan(\tanh x) = \text{sech } 2x$

$$\begin{aligned}
 \frac{d}{dx} \arctan(\tanh(x)) &\stackrel{\text{chain}}{=} \frac{1}{1+(\tanh(x))^2} \cdot (\tanh(x))' \\
 &= \frac{1}{1+(\tanh(x))^2} \cdot \text{sech}^2(x) \\
 &\stackrel{\text{Recall:}}{=} \frac{1 - (\tanh x)^2}{1 + (\tanh x)^2} = \frac{1 - \left(\frac{\sinh x}{\cosh x}\right)^2}{1 + \left(\frac{\sinh x}{\cosh x}\right)^2} = \frac{\cosh^2 x - \sinh^2 x}{\cosh^2 x + \sinh^2 x} \\
 &\stackrel{\text{Recall:}}{=} \frac{1}{\cosh 2x} = \text{sech } 2x
 \end{aligned}$$

Recall: $\cosh(x+y) = \cosh x \cosh y + \sinh x \sinh y$
 $\cosh^2 x - \sinh^2 x = 1$

Recall: $\text{sech}^2(x) = 1 - (\tanh x)^2$